



4 PUBLIC FACILITIES

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Kissimmee 2040 Comprehensive Plan

Public Facilities

prepared for

City of Kissimmee

prepared by

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date

November 2018

Cover photo source: City of Kissimmee Public Information Office

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GOALS, OBJECTIVES, AND POLICIES

This section stipulates goals, objectives, and implementing policies for the Public Facilities Element pursuant to §163.3177(6)(c), Florida Statutes.

CHAPTER 4A: POTABLE WATER SUB-ELEMENT

GOAL 4.1: PROTECTION OF THE POTABLE WATER SUPPLY.

The City of Kissimmee shall work with Tohopekalgia Water Authority (TWA) and the South Florida Water Management District (SFWMD) to ensure the protection of the potable water supply and delivery of safe and adequate potable water service. The City will ensure the service is managed in a cost effective manner to accommodate existing and future development.

OBJECTIVE 4.1.1: WATER QUALITY.

The City shall coordinate with the Tohopekalgia Water Authority (TWA), the provider of the City's water supply and responsible party managing the city's water supply, and shall cooperate with the Environmental Protection Agency (EPA) and the Florida Department of Environmental Protection (FDEP) in concert with the South Florida Water Management District to ensure all applicable water quality standards are met.

POLICY 4.1.1.1: WELLHEAD PROTECTION AREA.

The City adopted wellhead protection regulations in the Land Development Code (LDC) for existing and future major public wellheads. The City shall revise these regulations for consistency, if the Tohopekalgia Water Authority (TWA) or the SFWMD establish their own standards.

POLICY 4.1.1.2: INDUSTRIAL AND INTENSIVE COMMERCIAL USES WITHIN WELLHEAD PROTECTION AREA (WPA).

The City shall prohibit the Industrial Future Land Use Map designations and intensive commercial uses, as well as hazardous waste storage sites, solid waste disposal sites, and single-lined petroleum storage tanks, within the wellhead protection area.

POLICY 4.1.1.3: POTABLE WATER TREATMENT PLANTS.

The City shall assist FDEP in the review of potable water treatment plants. No new potable water treatment plants shall be allowed unless all applicable sections of the Florida Administrative Code (F.A.C.) governing such plants are met. Connections to existing potable water treatment plants not in compliance with applicable F.A.C. regulations shall also not be allowed. Such systems shall also comply with the pertinent rules and regulations



of any water supplier that may have jurisdiction. Land use impacts arising from the permitting of new potable water treatment plants, as well as expansions of existing plants, shall be consistent with the applicable provisions of this Plan prior to issuance of a development permit.

OBJECTIVE 4.1.2: COORDINATION OF POTABLE WATER FACILITIES AND ADEQUATE SUPPLY.

The City shall coordinate with the Tohopekaliga Water Authority (TWA), the sole supplier of central water systems within the City's jurisdictional boundaries, to ensure that the adopted potable water level of service standards will be met.

POLICY 4.1.2.1: LEVEL OF SERVICE (LOS) STANDARDS.

The areas of the City served by central potable water facilities shall meet the levels of service standards for potable water supply as specified in Chapter 8: Capital Improvement Element Policy 8.1.4.1.

POLICY 4.1.2.2: WATER SUPPLY CONCURRENCY.

Prior to the issuance of a development permit, the City shall consult with the Tohopekaliga Water Authority (TWA) to determine whether adequate water supplies to serve the proposed development or redevelopment project will be available no later than the issuance of a certificate of occupancy, or its functional equivalent.

POLICY 4.1.2.3: CENTRAL WATER SYSTEMS WITHIN THE CITY LIMITS.

The City shall require new development within the City limits to connect to central potable water systems. Existing development within the City that is not connected to a central potable water system must do so within one year of service becoming available. The City shall define availability as the physical presence of potable water facilities and infrastructure adjacent to subject property.

POLICY 4.1.2.4: EXISTING CUSTOMERS OF PRIVATE WATER SYSTEMS.

Existing customers of private water systems shall be connected to public central systems if the existing system falls into non-compliance with FDEP regulations, and/or the existing system is required to connect by consent order.

POLICY 4.1.2.5: NEW POTABLE WATER TREATMENT PLANTS.

New potable water treatment plants, as well as expansions of existing plants, shall be consistent with the Future Land Use Element (FLUE).

POLICY 4.1.2.6: CENTRAL POTABLE WATER SYSTEMS.

Consistent with the FLUE, the extension of central potable water systems shall be prohibited, unless the City determines that this service is needed for one of the following reasons:

1. A public health hazard exists for existing development, and extension will not serve as the basis for new development; or
2. When the facilities are extended to protect environmentally sensitive areas from impacts resulting from existing development.

POLICY 4.1.2.7: NON-POTABLE WATER USES.

The City will continue to coordinate with and support utility providers and governmental entities in requiring all new development to connect non-potable water uses, such as for irrigation, to available non-potable water systems, such as reuse systems.

POLICY 4.1.2.8: DUAL TRANSMISSION AND DISTRIBUTION LINES.

The City shall continue to coordinate with and support the Tohopekaliga Water Authority (TWA) in requiring new developments to install dual transmission and distribution lines to serve those developments where non-potable water is available or expected to be available.

POLICY 4.1.2.9: STORMWATER AS NON-POTABLE WATER SOURCE.

The City will continue to investigate the feasibility of stormwater as a non-potable water source when new stormwater management systems are permitted, and the City will coordinate with the utility providers and governmental entities in implementing feasible projects.

POLICY 4.1.2.10: ALTERNATIVE WATER SUPPLIES.

The City will coordinate with and support utility providers and the South Florida Water Management District in developing alternative water supplies, by providing incentives, and permitting assistance.

OBJECTIVE 4.1.3: WATER CONSERVATION.

The City shall annually review its Comprehensive Plan policies and Land Development Code regulations to address the conservation of water supplies and aim to reduce water demands by 5 percent proportionately in relation to growth by the year 2025.



POLICY 4.1.3.1: WATER SUPPLY CONSERVATION TECHNIQUES AND PROGRAMS.

Water supplies within the City's jurisdictional limits shall be conserved through the implementation of water conservation techniques and programs. Such techniques and programs should include, but are not limited to the following:

1. The City shall require the installation of water conserving plumbing fixtures in new or renovated construction.
2. The City shall continue to promote the efficient use of potable water and non-potable water in irrigation and landscaping practices through regulations in the Land Development Code.
3. The City will coordinate with the Osceola County Soil and Water Conservation District, the County Extension Office and the South Florida Water Management District to educate the public on the importance of water conservation and strategies for conserving water.
4. The City shall coordinate with the Tohopekaliga Water Authority (TWA) to promote the use of water conserving techniques, including through the use of a progressive rate structure.
5. The City shall cooperate with emergency water conservation measures of the South Florida Water Management District, and the Tohopekaliga Water Authority.
6. The City shall use Florida Water Star program criteria for the construction of new single-family residences.
7. Through the land development code, the City shall continue to place restrictions on irrigation during periods of time of critical concern that are consistent with 40E-24 of the Florida Administrative Code.

OBJECTIVE 4.1.4: COORDINATION WITH WATER SUPPLIER.

The City shall coordinate with the Tohopekaliga Water Authority (TWA) and South Florida Water Management District (SFWMD) to conserve and manage potable water resources, and maximize the use of existing facilities throughout its jurisdiction. The City shall meet this objective in order to discourage urban sprawl.

POLICY 4.1.4.1: WATER SUPPLIERS; AUTHORITY OF FDEP.

The planning of potable water facilities within the City shall be pursuant to the ordinances and rules of the City, created pursuant to Section 125.01 and Chapter 153, F.S., and as

they may be, from time to time, amended, supplemented or otherwise modified. The City shall coordinate such efforts with the Tohopekaliga Water Authority (TWA). Neither revisions to the enabling legislation of the TWA, nor mechanisms implemented by the City, shall conflict with the exclusive authority of the FDEP and the South Florida Water Management District (SFWMD) to regulate the consumptive use of water as provided under Chapter 373, F.S.

POLICY 4.1.4.2: WATER SUPPLIER MASTER PLAN.

The City, in concert with Osceola County, shall coordinate with the SFWMD in the maintenance of the Regional Water Supply Plan 2015 and shall support the following:

1. Identify the sustainable quantities of traditional groundwater sources available for water supplies that can be used without causing harm to the water resources and associated natural systems.
2. Identify water conservation and alternative water supply development options to meet reasonable-beneficial water demands that are in excess of the sustainable yield of traditional groundwater sources.
3. Protect and enhance the environment, including the natural resource areas and systems identified by the Districts as well as any federal, state, and locally identified natural resource areas.
4. Provide information to support local government comprehensive plans.
5. Achieve compatibility and integration with other state and federal regional resource initiatives.
6. Establish consistent regulatory programs to accomplish the above.

POLICY 4.1.4.3: WATER SYSTEM FACILITY CONSTRUCTION.

The construction of water system facilities and associated infrastructure within the City, except for facilities which serve only the agricultural needs of the property on which the system is located, shall be regulated by the City.

POLICY 4.1.4.4: POTABLE WATER FACILITY EVALUATION CRITERIA.

The City shall examine the capital improvement projects proposed by the Tohopekaliga Water Authority (TWA) in their Capital Budget, and shall provide input regarding project ranking, based on the following priority level guidelines:

1. "Level 1": Whether the project is needed to:



- a. Protect public health, safety, and environmentally sensitive natural resources.
 - b. Fulfill the City's legal commitment to provide facilities and services.
 - c. Preserve or achieve full use of existing facilities.
2. "Level 2": Whether the project accomplishes the following:
 - a. Increases efficiency of existing facilities.
 - b. Prevents or reduces future improvement costs.
 - c. Provides service to developed areas lacking full services or promotes in-fill development.
 3. "Level 3": Whether the project:
 - a. Represents a logical extension of facilities and services in a manner consistent with Future Land Use Element goals, objectives and policies, including the Future Land Use Map.

POLICY 4.1.4.5: WATER MANAGEMENT DISTRICT PERMITS REQUIRED.

Construction of any water system facility, including any extraction mechanisms shall require the issuance of all required permits including any required consumptive use permit (CUP) from the South Florida Water Management District pursuant to Chapter 373, F.S. The City's approval shall be conditioned upon issuance of such permits prior to the commencement of construction.

OBJECTIVE 4.1.5: REGIONAL COORDINATION OF WATER RESOURCES.

The City shall be proactive in regional water resources planning efforts, through enhanced coordination with adjacent jurisdictions, the Tohopekaliga Water Authority (TWA), the South Florida Water Management District, and all water supply coordinating entities.

POLICY 4.1.5.1: DIMINUTION OR DEPLETION OF ESSENTIAL WATER SOURCES.

Consistent with applicable State law and rules, any land use posing a threat to the City's long-term natural resource base through the diminution, contamination, or depletion of essential water resources shall not be permitted. "Diminution" and "depletion" shall not include the diminution or depletion of water resources due to the consumptive use of water.

POLICY 4.1.5.2: POTABLE WATER DEMAND MAY NOT EXCEED CONSUMPTIVE USE PERMIT (CUP).

The City, through direct coordination with the Tohopekaliga Water Authority (TWA), as well as the FDEP, and the South Florida Water Management District, shall ensure potable water demands caused by development do not exceed the consumptive use permits (CUP) held by TWA.

POLICY 4.1.5.3: ALTERNATIVE WATER SOURCES.

The City shall, with the Tohopekaliga Water Authority (TWA), coordinate with the South Florida Water Management District to find alternative sources of potable water.

POLICY 4.1.5.4: LEVEL OF SERVICE (LOS) STANDARDS MAINTAINED.

In order to adequately serve the existing and future population, the City shall proactively work with the Tohopekaliga Water Authority (TWA) and the South Florida Water Management District to ensure adopted level of standards are met and any identified facility deficiencies are corrected.

OBJECTIVE 4.1.6: TEN YEAR WATER SUPPLY FACILITIES WORK PLAN.

The City shall maintain a Water Supply Facilities Work Plan (Work Plan) for water supply projects within 18 months of the regional water supply Master Plan update.

POLICY 4.1.6.1: WORK PLAN INCORPORATED INTO COMPREHENSIVE PLAN.

The Work Plan, included in the Capital Improvement Plan of the Capital Improvement Element, shall provide projects identified by the City, and through coordination with the Tohopekaliga Water Authority (TWA) and the South Florida Water Management District, that are required to provide adequate potable water supplies to meet existing and future demand. These tables will be updated annually to reflect the adopted capital improvements programs of the TWA.

POLICY 4.1.6.2: AGENCY COOPERATION TO MAINTAIN WORK PLAN.

Maintaining and amending the Work Plan shall be a cooperative effort between the City, the Tohopekaliga Water Authority (TWA), and the South Florida Water Management District.

POLICY 4.1.6.3: CONSISTENCY WITH REGIONAL PLANS.

The Work Plan shall be consistent with the most current Regional Water Supply Plan of the South Florida Water Management District.



POLICY 4.1.6.4: COORDINATION OF FUTURE UPDATES TO WORK PLAN.

The updates to the Work Plan shall be consistent with the most current water supply plans of the Tohopekaliga Water Authority (TWA).

POLICY 4.1.6.5: UPDATES BASED ON LEVEL OF SERVICE STANDARDS.

The updates to the Work Plan shall evaluate resources based upon the potable water levels of service (LOS) standards established in the Capital Improvements Element.

POLICY 4.1.6.6: UPDATES TO ADDRESS WATER SUPPLY, CONSERVATION AND REUSE.

The updates to the Work Plan shall identify feasible traditional and alternative water supply sources, including water conservation and reuse that are being utilized to meet existing and projected water demands.

POLICY 4.1.6.7: WORK PLAN IS BASIS FOR PRIORITIZATION OF FUTURE SYSTEM IMPROVEMENTS.

The Work Plan shall be used to prioritize and coordinate the expansion and improvement of facilities used to withdraw, transmit, treat, store, and distribute potable water to meet current and future needs within the City.

CHAPTER 4B: WASTEWATER SUB-ELEMENT

GOAL 4.2 WASTEWATER FACILITIES.

Ensure an effective system of wastewater facilities to promote the health, safety, and welfare of City residents and the natural environment in a manner that protects investments in existing facilities, promotes orderly, compact growth, and meets existing and projected demand.

OBJECTIVE 4.2.1 ENSURE WASTEWATER CAPACITY.

The City of Kissimmee shall adopt procedures to require that at the time a development permit is issued, adequate facility capacity is available or will be available when needed to serve the development.

POLICY 4.2.1.1 LEVEL OF SERVICE STANDARDS.

The level of service standards as established in Chapter 8: Capital Improvement Element Policy 8.1.4.1 for wastewater facilities within the City shall be used as the basis for determining availability of facility capacity and demand generated by a development;

POLICY 4.2.1.2: COMPLIANCE WITH LEVEL OF SERVICE STANDARDS.

The City shall coordinate with the Tohopekalgia Water Authority (TWA) regarding improvements for replacement, expansion or increase in capacity of facilities so that demand is compatible with the adopted level of service standards for the facilities. Issuance of development orders and permits shall be conditioned upon the submission of plans that are in compliance with the City's level of service standards. Prior to issuance of development orders or permits the project shall provide the City copies of all applicable Federal, State, and local permits for wastewater facilities.

POLICY 4.2.1.3: COORDINATION OF INFORMATION.

The City of Kissimmee shall coordinate with the Tohopekalgia Water Authority (TWA) to develop procedures for updating facility demand and capacity information and the preparation of an annual summary of capacity and demand

POLICY 4.2.1.4: COORDINATE LAND USE IMPACTING WASTEWATER SYSTEMS.

Realizing the substantial impacts of the location and timing of development as well as the density or intensity of specific land use activities on urban service delivery systems, the City of Kissimmee shall coordinate land uses with the County and the Tohopekalgia Water Authority (TWA) in order to effectively monitor water consumption for wastewater uses pursuant to the terms of the water and wastewater Interlocal Agreement established between Osceola County, the City of Kissimmee, Orange County, and TWA in November 2004.

POLICY 4.2.1.5: WASTEWATER SERVICE AREAS.

Provision of centralized wastewater service shall be provided to the urban service area within the City's jurisdictional area as provided by the Tohopekalgia Water Authority (TWA).

POLICY 4.2.1.6: EXPANSION OF WASTEWATER TREATMENT FACILITIES.

The City of Kissimmee shall coordinate with the Tohopekalgia Water Authority (TWA) regarding expansion of the capacity of its wastewater treatment facilities.

POLICY 4.2.1.7: MANDATORY CONNECTION.

All new development within the City's limits will be required to connect to central wastewater facilities.

OBJECTIVE 4.2.2: MAINTAINING A SCHEDULE OF WASTEWATER CAPITAL IMPROVEMENTS.

The City shall coordinate with the Tohopekalgia Water Authority (TWA) to maintain a five-year schedule of capital improvements for public facilities, specifically wastewater facilities.



POLICY 4.2.2.1: CAPITAL IMPROVEMENT SCHEDULE.

The City Commission shall coordinate with the Tohopekaliga Water Authority (TWA) to include an annual evaluation of capital improvement projects proposed for inclusion in the five-year schedule of capital improvements.

POLICY 4.2.2.2: WASTEWATER FACILITY EVALUATION CRITERIA.

The City shall examine the capital improvement projects proposed by the Tohopekaliga Water Authority (TWA), and shall provide input regarding project ranking, based on the following priority level guidelines:

1. "Level 1": Whether the project is needed to:
 - a. Protect public health, safety, and environmentally sensitive natural resources.
 - b. Fulfill the City's legal commitment to provide facilities and services.
 - c. Preserve or achieve full use of existing facilities.
2. "Level 2": Whether the project accomplishes the following:
 - a. Increases efficiency of existing facilities.
 - b. Prevents or reduces future improvement costs.
 - c. Provides service to developed areas lacking full services or promotes in-fill development.
3. "Level 3": Whether the project:
 - a. Represents a logical extension of facilities and services in a manner consistent with Future Land Use Element goals, objectives and policies, including the Future Land Use Map.

OBJECTIVE 4.2.3: ON-SITE WASTEWATER TREATMENT SYSTEMS.

Where central wastewater system facilities are not available, the City shall permit a private wastewater disposal system pursuant to conditions stipulated in §10-5-54, Code of Ordinances, as hereafter may be amended. The City shall coordinate with the County Health Department and the Florida Department of Health programs to assist in implementation of State regulations imposing mandated standards for inspections, operation, and maintenance of on-site wastewater treatment systems.

POLICY 4.2.3.1: USE OF ON-SITE WASTEWATER TREATMENT SYSTEMS.

Private, on-site treatment and disposal systems shall, at a minimum, meet the requirements set forth by the Department of Health in Chapter 64E-6, Florida Administrative Code, as amended. When central wastewater becomes available, connection is required by ordinance.

POLICY 4.2.3.2: CONDITIONS GOVERNING DEVELOPMENT ORDERS OR PERMITS.

Issuance of development orders or permits shall be conditional upon demonstrated compliance with applicable Federal, State, and local regulations governing on-site wastewater treatment systems.

CHAPTER 4C: AQUIFER RECHARGE SUB-ELEMENT

GOAL 4.3: PROTECT FUNCTIONS OF GROUNDWATER AQUIFER RECHARGE AREAS.

The functions of natural groundwater aquifer recharge areas and natural drainage features within the City, including their water quality and water quantity, shall be protected and maintained.

OBJECTIVE 4.3.1: COORDINATE EFFORTS TO PROTECT AQUIFER RECHARGE AREAS.

The City shall cooperate and coordinate with Osceola County, the Tohopekaliga Water Authority (TWA), the City of St. Cloud, the Reedy Creek Improvement District, and the South Florida Water Management District, to ensure proper maintenance of aquifer recharge area functions and to determine whether new protection programs or additional coordination is needed to achieve aquifer protection objectives.

POLICY 4.3.1.1: PROTECT SURFICIAL AQUIFER RECHARGE AREAS AND GROUNDWATER QUALITY.

The City currently has no “prime” surficial aquifer recharge areas identified by the South Florida Water Management District. The City shall assist in protecting groundwater from point and non-point pollution sources. Similarly, the City shall promote the conservation and efficient use of water as it travels through groundwater systems, and shall promote maintenance of adequate supplies of high quality groundwater.

POLICY 4.3.1.2: COORDINATE GROUNDWATER MONITORING.

The City shall assist the State and South Florida Water Management District in managing water quality by assisting in preventing the discharge of inadequately treated wastewater and poor quality stormwater into public water bodies and aquifer recharge areas, and shall provide best available data to the South Florida Water Management District, and the Florida Department of Environmental Protection, to ensure that groundwater resources



are monitored for point and nonpoint source pollution affecting the Floridan, and Surficial aquifers.

POLICY 4.3.1.3: DEEP AQUIFER WATER CONSERVATION.

In order to protect the quality and quantity of deep aquifer water resources, the City shall coordinate with the Tohopekaliga Water Authority (TWA), and the South Florida Water Management District and other applicable regulatory agencies in identifying free flowing artesian aquifer wells and in requiring corrective measures, including capping, plugging, or installing regulatory devices that control the discharge of water from the deep aquifer.

POLICY 4.3.1.4: RETAIN STORMWATER RUN-OFF TO MAXIMIZE AQUIFER RECHARGE.

The City shall continue to enforce stormwater management regulations requiring retention of stormwater run-off to maximize groundwater recharge.

POLICY 4.3.1.5: COORDINATE WITH OTHER RECHARGE PROTECTION PROGRAMS.

The City will coordinate with local, state, and federal agencies to achieve regional aquifer recharge protection objectives, especially protection of prime recharge areas located in northwest Osceola County.

POLICY 4.3.1.6: FDEP GROUNDWATER CONTAMINATION LIST.

The City shall coordinate with Osceola County and the Tohopekaliga Water Authority (TWA) to monitor the Florida Department of Environmental Protection (FDEP) Site Investigation Section list for groundwater contamination possibilities, and will work cooperatively with FDEP to resolve any conditions impacting groundwater supplies regarding potential sites.

OBJECTIVE 4.3.2: PLANNING FOR THE PROTECTION OF AQUIFER RECHARGE AREAS.

To meet current and future demands for potable water resources and to protect natural resources, the City shall cooperate with other governmental agencies, especially the South Florida Water Management District (SFWMD), to ensure existing and projected development do not adversely affect groundwater resources.

POLICY 4.3.2.1: AQUIFER RECHARGE PROTECTION MEASURES.

Recognizing that aquifer recharge areas, groundwater resources, wellhead protection areas, and that lakes, rivers, creeks, and wetlands within the City are interconnected systems vital to the protection of water resources, aquifer recharge protection measures for “high” recharge areas, if identified, shall be consistent with the Conservation and Future Land Use Elements, as well as this element of the Comprehensive Plan.

POLICY 4.3.2.2: AQUIFER RECHARGE PROTECTION MECHANISMS.

The City may use open space designations, land acquisition strategies, planned development zoning districts, conservation easements, or cooperative management agreements to protect identified aquifer recharge areas.

POLICY 4.3.2.3: HAZARDOUS WASTE STORAGE FACILITIES.

The City shall regulate the location of hazardous waste storage facilities, petroleum storage tanks, or any other adverse land use within “high” aquifer recharge areas, if identified, through the Land Development Code.

CHAPTER 4D: STORMWATER MANAGEMENT SUB-ELEMENT

GOAL 4.4: STORMWATER MANAGEMENT FACILITIES.

Ensure the provision of a system of stormwater management facilities in a manner that protects persons and property from flooding, prevents negative impacts to the surficial groundwater aquifer, and safeguards surface waters against the degradation of water quality, to promote the health, safety, and welfare of City residents and the natural environment.

OBJECTIVE 4.4.1: STORMWATER CAPACITY AND QUALITY.

The City of Kissimmee shall maintain regulations for the adequate provision of stormwater facilities needed to serve existing and proposed development.

POLICY 4.4.1.1: LEVEL OF SERVICE STANDARDS.

The level of service standards established in Capital Improvements Element Policy 8.1.4.1 shall be used as the basis for determining the availability of facility capacity and the demand generated by a development prior to the issuance of a development permit.

POLICY 4.4.1.2: STORMWATER QUALITY.

All stormwater discharge within the City shall meet all State water quality standards, including those contained in Chapter 62-25, Florida Administrative Code, as well as any Total Maximum Daily Load thresholds that may be established for Lake Tohopekaliga, Shingle Creek, Mill Slough, or other surface waterbodies located within the City.

POLICY 4.4.1.3: ANNUAL REPORT OF FACILITY PERFORMANCE.

The City of Kissimmee shall compile data regarding existing stormwater facilities and/or management areas in the City, annually, and shall provide this data to Osceola County for coordination activities.



OBJECTIVE 4.4.2: STORMWATER MANAGEMENT IMPROVEMENTS.

The City shall identify and implement stormwater management projects and programs necessary to meet existing needs and projected demands.

POLICY 4.4.2.1: CAPITAL IMPROVEMENT SCHEDULE.

The City Commission shall annually evaluate and rank capital improvement projects proposed for inclusion in the five-year schedule of capital improvement needs.

POLICY 4.4.2.2: STORMWATER FACILITY EVALUATION CRITERIA.

Proposed capital improvement projects shall be evaluated and ranked according to the following priority level guidelines:

1. "Level 1": Whether the project is needed to:
 - a. Protect public health, safety, and environmentally sensitive natural resources.
 - b. Fulfill the City's legal commitment to provide facilities and services.
 - c. Preserve or achieve full use of existing facilities.
2. "Level 2": Whether the project accomplishes the following:
 - a. Increases efficiency of existing facilities.
 - b. Prevents or reduces future improvement costs.
 - c. Provides service to developed areas lacking full services or promotes in-fill development.
3. "Level 3": Whether the project:
 - a. Represents a logical extension of facilities and services in a manner consistent with Future Land Use Element goals, objectives and policies, including the Future Land Use Map.

POLICY 4.4.2.3: CENTRALIZED STORMWATER MASTER PLANS.

The City shall encourage the creation, implementation, and maintenance of central stormwater facility master plans in order to encourage development and redevelopment activities within designated areas. These areas shall include, at a minimum, the Community Redevelopment Area (CRA) District, and the Vine Street Corridor.

POLICY 4.4.2.4: STORMWATER UTILITY FUND.

The City shall fund projects to mitigate identified deficiencies through the Stormwater Utility, while impacts associated with new development or redevelopment projects must be mitigated by the developer in accordance with adopted level of service standards and/or central master plans. Eligible projects shall include the production of centralized stormwater master plans.

POLICY 4.4.2.5: ADDITIONS OF STORMWATER MANAGEMENT FACILITY PROJECT APPROVALS.

All required federal, State, and County permits shall be obtained before the City undertakes or authorizes contractors to undertake construction and/or operation of facilities.

OBJECTIVE 4.4.3: PROTECT NATURAL DRAINAGE FEATURES.

The City shall manage and coordinate its stormwater review and implementation process to meet existing and future needs, while protecting the quality and functions of natural drainage features and surface waterbodies.

POLICY 4.4.3.1: NATURAL STORAGE CAPACITY OF WATERSHEDS.

The City shall:

- Promote the ecological, biological, and hydrological role of surface waters in aquifer recharge and in supporting surface vegetation.
- Manage the location, design and intensity of urban development in order to foster continuance of natural hydrological processes, including preserving recharge areas, promoting on-site retention of surface waters and natural return of surface water into the soil.
- Require the integration of natural storage areas and natural drainage courses into water management plans for new development.
- Require that applicants for new development consider use of isolated wetlands for flood attenuation as may be appropriate in the design of drainage and surface water management plans.

POLICY 4.4.3.2: MAINTAIN REGULATIONS FOR LAND AND WATER MANAGEMENT.

The City of Kissimmee shall promote both land and water management programs and practices which retard runoff and enhance percolation in order to increase the quantity and protect the quality of ground and surface water. Land use controls, such as subdivision regulations, zoning, including site plan review and performance criteria, and special erosion control, landscape and flood management requirements shall



be maintained in the Land Development Code and periodically updated based on improved knowledge of problems, issues, and best management practices and low impact development techniques.

POLICY 4.4.3.3: PURSUE THE DEVELOPMENT OF ADEQUATE OFF-SITE SURFACE WATER MANAGEMENT FACILITIES.

The City shall manage stormwater based on watershed management plans. Implementing strategies shall provide a basis for evaluating the performance of existing off-site drainage facilities, identifying existing and potential future problems or issues, and funding necessary structural and non-structural system improvements for effective surface water management.

POLICY 4.4.3.4: COORDINATE WATERSHED MANAGEMENT PLANS AND POLICIES WITH APPROPRIATE PUBLIC AGENCIES.

The City shall assure coordination of watershed management plans and policies with appropriate local, regional, state and federal agencies, including Osceola County, the South Florida Water Management District, the East Central Florida Regional Planning Council, the Florida Department of Environmental Protection, the Agricultural Extension Service, the United States Army Corps of Engineers, and other appropriate agencies.

POLICY 4.4.3.5: BUFFER ZONE REQUIREMENTS.

The City shall continue to include buffer zone requirements for wetland areas in the Land Development Code, and be consistent with the Conservation Element of this Comprehensive Plan. Wetland buffer distances shall be at least 50 feet. There shall be no development within the Floodway as established by FEMA, except for minor improvements such as boat docks, where certification and supporting technical documents provided by a registered professional engineer demonstrate that the encroachment shall not result in any increase in flood levels during occurrence of the base flood discharge.

POLICY 4.4.3.6: MANAGING LAND USE IN THE FLOODPLAIN.

The City shall continue to retain stormwater management and floodplain regulations that shall address necessary restrictions on encroachment, alteration, and compatible use of the 100-year floodplain and major drainage corridors, including Shingle Creek, and Mill Slough.

POLICY 4.4.3.7: PROMOTION OF LOW-IMPACT DEVELOPMENT TECHNIQUES.

The City shall encourage the use of low-impact development techniques for all land development and redevelopment projects. Low-impact development involves structural and non-structural techniques designed to mimic natural stormwater flows and improve

stormwater runoff quality. Techniques may include green roofs, rain gardens, constructed wetlands, infiltration swales, and open space conservation.

POLICY 4.4.3.8: INSPECTION AND MAINTENANCE OF DRAINAGE SYSTEMS.

As part of the master stormwater drainage implementation program, the City shall ensure that major drainage systems receive periodic inspections, including appropriate maintenance activities at least once a year.

POLICY 4.4.3.9: NATIONAL POLLUTION DISCHARGE ELIMINATION SYSTEM (NPDES).

The City of Kissimmee shall maintain its stormwater control measures in accordance with regulations and requirements associated with the National Pollution Discharge Elimination System (NPDES) permit, and shall renew the permit as required by federal regulations.

POLICY 4.4.3.10: PUBLIC EDUCATION.

The City shall provide information, as feasible, to educate the public in the importance of minimizing pollution to the City's stormwater management systems. The programs may include: newsletters, website updates, community meetings, charrettes, public hearings and other public information, community visioning, commissions and standing advisory committees, stormwater task forces or other special committees.

CHAPTER 4E: SOLID WASTE SUB-ELEMENT

GOAL 4.5: SOLID WASTE FACILITIES.

Ensure availability of needed solid waste facilities in a manner that protects investments in existing facilities and promotes orderly, compact growth, and meets existing and projected demand.

OBJECTIVE 4.5.1: ENSURE CAPACITY TO MEET PROJECTED DEMAND.

The City of Kissimmee shall maintain procedures to require adequate facility capacity is available, or will be available, when needed to meet projected demand. In the case of individual development projects, capacity shall be available to serve the development, at the time a development permit is issued.

POLICY 4.5.1.1: LEVEL OF SERVICE STANDARDS.

Solid waste generation rates identified in Capital Improvements Element Policy 8.1.4.1 shall serve as the City's level of service standards for solid waste facilities. These figures shall be used as the basis for determining the availability of solid waste facility capacity, and they only include the portion of the waste stream that is not diverted for recycling purposes.



POLICY 4.5.1.2: COMPLIANCE WITH LEVEL OF SERVICE STANDARDS.

Issuance of development orders and permits within the City shall be conditioned upon the submission of plans which are in compliance with the City's level of service standards. The City shall continue to include an adequate facilities requirement as part of the Land Development Code.

POLICY 4.5.1.3: STUDY LEVEL OF SERVICE STANDARD.

The City shall coordinate with Osceola County to develop a level of service standard that reflects available capacity.

POLICY 4.5.1.4: REDUCTION OF SOLID WASTE VOLUMES.

The City shall work with Osceola County and other organizations to reduce the volume of solid waste through recycling activities, educational efforts and programs, and by assisting in the maintenance of drop-off collection sites in the County.

POLICY 4.5.1.5: DEVELOPMENT OF NEW FACILITIES.

The City shall assist Osceola County in planning for the timely provision of needed solid waste facilities in accordance with applicable regulations.

OBJECTIVE 4.5.2.: COORDINATION ACTIVITIES.

The City shall assure that deficiencies in solid waste facilities are avoided through coordination with Osceola County's solid waste management program.

POLICY 4.5.2.1: GENERAL INTERGOVERNMENTAL COORDINATION.

The City shall cooperate with County officials and technicians on a Joint Committee or through other appropriate mediums in order to address county-wide or regional approaches for achieving access to resource recovery facilities or other alternatives to conventional landfill operations. Other specific issues which shall be addressed in coordination with Osceola County include:

- Enhancing solid waste collection and transfer operations;
- Management strategies for implementing recycling efforts;
- Curbing illegal dumping of solid waste as well as disposal activities which adversely impact natural systems;
- Developing improved information dissemination regarding hazardous waste generators;
- Determining feasibility of hazardous waste storage/transfer facilities;

- Improving management of the collection and disposal of hazardous waste;
- Drafting policy for appropriate regulatory measures governing solid waste and hazardous waste including identification of long term operating costs and capital improvement needs associated with various policy options.

POLICY 4.5.2.2: INFORMATION COORDINATION.

The City of Kissimmee shall coordinate with the County, and with franchised waste collection operators within the City regarding demand and capacity information, and shall prepare annual summaries of capacity and demand information for solid waste collection and disposal activities.

POLICY 4.5.2.3: COORDINATE WITH OSCEOLA COUNTY SOLID WASTE AND HAZARDOUS WASTE PLANNING EFFORTS.

The City shall offer assistance to Osceola County as the County assesses and plans for solid waste and hazardous waste management in a manner consistent with the provisions of §403.7265, F.S.

POLICY 4.5.2.4: EDUCATIONAL OUTREACH.

The City shall coordinate with Osceola County in the distribution of educational materials to the public concerning the proper storage, handling, collection, transport, and disposal of waste oil, batteries and household chemical waste.

OBJECTIVE 4.5.3: FINANCIAL FEASIBILITY.

The City shall provide financial resources to support solid waste operations and will include these expenditures in the Five-Year Schedule of Capital Improvements of the Capital Improvements Element.

POLICY 4.5.3.1: CAPITAL IMPROVEMENTS.

Capital improvements for solid waste disposal shall be identified and prioritized on an annual basis within the Capital Improvements Element.

POLICY 4.5.3.2: SOLID WASTE FACILITIES.

Construction and operation of solid waste facilities will be funded with revenue sources that may include user fees, improvement fees, impact fees, franchise fees, bonds and grant from state and federal programs, and other identified fees.

POLICY 4.5.3.3: RATE SCHEDULE.

The City shall maintain a solid waste rate schedule that will provide sufficient funds to adequately serve both residents and businesses.



POLICY 4.5.3.4: ANNUAL REVIEW OF RATE SCHEDULE.

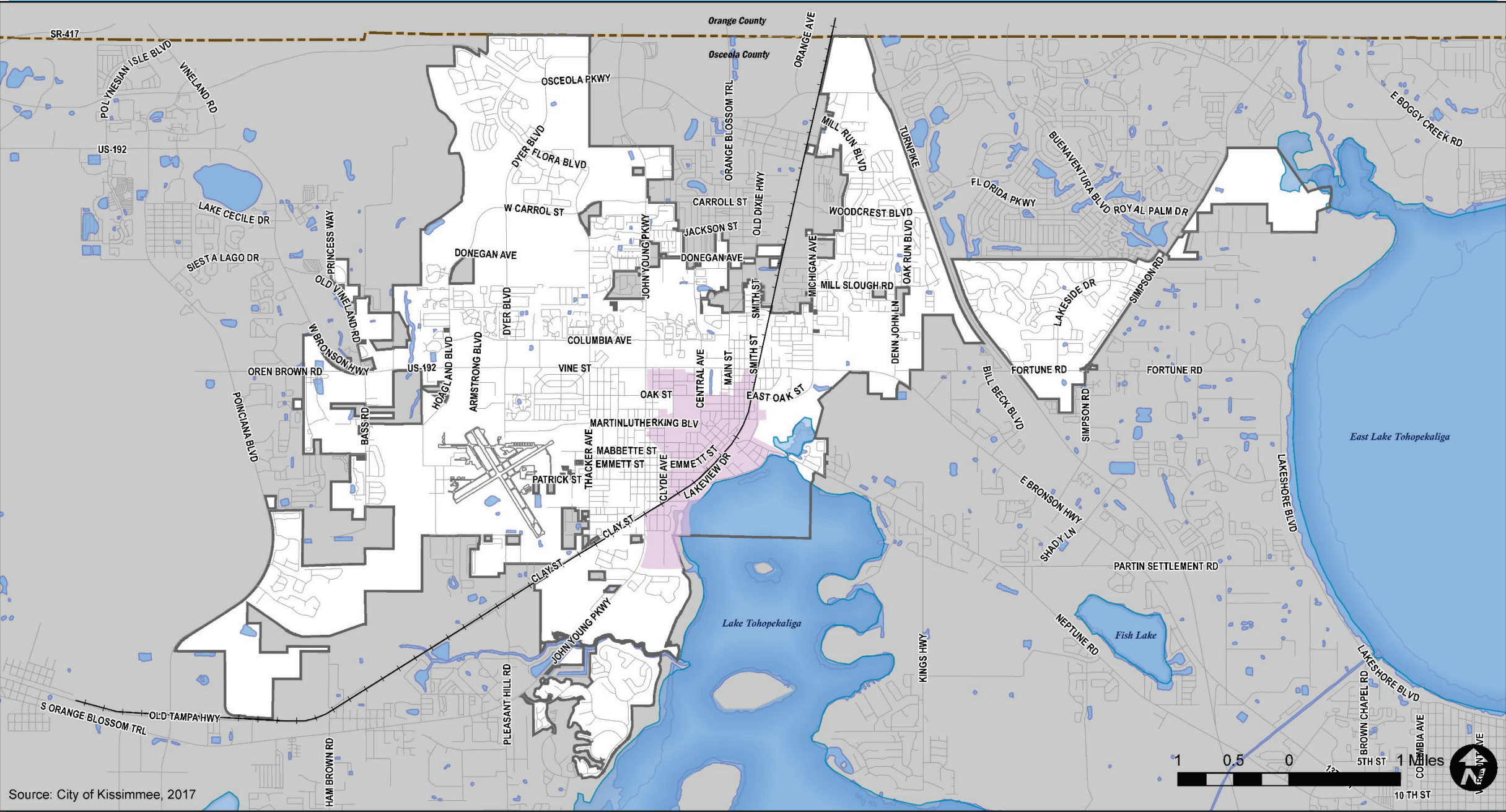
The City shall review the solid waste rate schedule annually, during the budget adoption process, and amend the rates as needed to fund solid waste operations.

MAPS

2040 Stormwater Master Plan Area

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Figure 4-1: 2040 Stormwater Master Plan Area



- Stormwater Master Plan Area
- City Limits
- County Boundary
- Unincorporated Area



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DATA, INVENTORY, AND ANALYSIS

This section provides the data inventory and analysis used to support the goals, objectives, and policies for the Public Facilities Element pursuant to §163.3177(6)(c), Florida Statutes.

CHAPTER 4A: POTABLE WATER SUB-ELEMENT

The data within this sub-element is based on the *City of Kissimmee Water Supply Facilities Work Plan dated January 2018*, which is incorporated by reference.

INTRODUCTION

The provision of potable water facilities is integral to ensuring the public health, safety, and welfare of a community. All local governments, even those that are not utility providers, must ensure that an adequate potable water supply is available to support future populations. The City of Kissimmee is addressing the future need for water resources through the protection of natural resources within its jurisdiction. Additionally, the City is partnering with state agencies to pursue water supply planning in a regional manner.

The City is not a provider of potable water service. The City was a provider until Osceola County and the City decided to find ways to develop an efficient and environmentally-sensitive approach to the comprehensive supply, distribution, and treatment of water. Therefore, the City, County, and area state representatives drafted a special bill creating Tohopekaliga Water Authority (TWA), which the Florida Legislature approved during the 2003 legislative session and was officially adopted on October 31, 2003. This consolidated governmental utility authority utilized existing resources to avoid duplication of effort, prevented the inefficient expansion of potable water facilities, and developed a cooperative regional approach to the provision of services. It should be noted that because the City is located completely within TWA's service area and within Osceola County, some of the following information is not completely specific to the City, because TWA's service area extends to the western border of the County.

POTABLE WATER FACILITIES AND SYSTEMS

A potable water supply system consists of a water source, treatment plants, and storage and distribution networks. Either surface or groundwater, or some combination of the two, constitute the supply source for a system. The selection of a source must consider the type and quality of source available, and the cost of developing the source for use. Before consumption, most water must be treated to remove impurities. After treatment, the water is supplied to individual users through a network of pipes and storage reservoirs. Large transmission lines, called distribution mains, carry water to major demand areas and interconnect with smaller lines, which eventually supply individual users. Water is delivered under pressure within the distribution system to ensure adequate flow meets customer demand. To



provide adequate quantities and pressure to meet peak use and fire flow demands, storage tanks are linked with the distribution system.

REGULATORY REQUIREMENTS

Two water management districts oversee Osceola County: the St. Johns River Water Management District (SJRWMD) and the South Florida Water Management District (SFWMD). The SJRWMD includes the eastern portion of the County along Brevard and Indian River Counties. The SFWMD encompasses the County's high-growth areas including the City of Kissimmee. TWA and the City must coordinate with each of the water management districts when addressing water resources protection, identifying water sources for public use, and meeting future state mandates. It should be noted that the City is located totally within the SFWMD's jurisdiction, and the below analysis only discusses that District.

The Florida Department of Environmental Protection (FDEP) has primary responsibility for the State Safe Drinking Water Program under Chapter 62-555, Florida Administrative Code (F.A.C.), by regulating public drinking water (potable water) systems, including:

- Community water systems serving 25 residents or more; and
- Non-community water systems serving 25 transient/non-residential persons daily.

FDEP permits new systems, monitors compliance, and enforces regulations. The Florida Department of Health (FDOH) administers regulatory programs for private water systems (wells), and all public water-supply systems not regulated by FDEP, by setting water-quality standards. In addition, FDOH has authority over the location of private wells if a septic tank is installed. Water-Use Permits for permission to extract water other than at domestic rates are issued by the SFWMD.

CREATION OF A REGIONAL UTILITY AUTHORITY

In late 2001, the City entered into an interlocal agreement with Osceola County and the City of St. Cloud to create a local government working group and authorize the use of professional services relating to a study to determine the feasibility of creating a regional water and wastewater authority. In late 2002, the City of St. Cloud decided to withdraw from the proposed authority. However, Osceola County and the City of Kissimmee continued with the creation of the regional utility authority and entered into an interlocal agreement in early 2003.

During the 2003 legislative session a special bill was passed that created the regional utility authority known as Tohopekaliga Water Authority (TWA). The Authority's boundary consists of all areas within Osceola County, excluding the legal boundaries of the Reedy Creek Improvement District and the boundaries authorized by law to be served or provided water and wastewater by the City of St. Cloud.

The transferring of the existing utility systems to TWA occurred through a transitional interlocal agreement. The City of Kissimmee's operational and administrative staff transferred to the operational and administrative staff of the TWA and all of Osceola County's right to acquire systems such as the Harmony system, the Poinciana system, and the Florida Water Services systems also transferred to the Authority.

TWA is a regional approach that will provide a more efficient and comprehensive means of providing water to the citizens of Osceola County and cease the disjointed and fragmented nature in which water has traditionally been provided in the County.

TOHOPEKALIGA WATER AUTHORITY

The Tohopekaliga Water Authority provides potable water to residents and businesses within the City of Kissimmee and to portions of Osceola County's unincorporated northwest area, as well as portions of Orange County's.

The TWA consists of three sub-areas and 14 water supply facilities, all of which draw their water from the Floridan Aquifer. Due to the high quality of groundwater from deep in this aquifer, the TWA treatment facilities need only provide chlorination, hydrogen sulfide removal, and pH adjustment in some areas prior to distribution.

SFWMD issued TWA a consumptive use permit (CU49-00103-W) that is set to expire on June 14, 2027. The permit allows an average potable water daily flow of 36.5 million gallons per day (MGD) and a maximum daily allocation of 51.86 MGD.

By the year 2013, the consumptive use permit required a minimum of 15 MGD of TWA's water supply be provided by alternative water supply (AWS) projects. Alternative water supply projects identified in the consumptive use permit include a Kissimmee River Basin project, and the proposed Cypress Lake Brackish Groundwater Wellfield. TWA's potable central water system currently provides an average of approximately 25 MGD to its customers. The combined capacity of the TWA water treatment plants is 56.6 MGD.

ANALYSIS OF FUTURE POTABLE WATER NEEDS

The City's growth management policy has been to encourage compact development in the existing public facility service area. In addition to this Element, the Future Land Use Element (FLUE) also outlines the issues involving provision of services within the City in order to manage growth and prevent urban sprawl. The City's regulations state that connection to central utilities is mandatory.

Mandatory connection to central potable water facilities requires providers of central utilities (in this case, TWA) to provide capacity in a timely fashion. Interlocal agreements with the TWA, the City of St. Cloud,



and the County will continue to be necessary to plan for future utility expansions. The primary responsibility of TWA is to promote the responsible, equitable and economical use of allocated water supplies.

The TWA is providing water supply from groundwater through the upper Floridan aquifer. City of Kissimmee is within the Toho I service area. As shown in Table 4-1, the Toho I (Kissimmee System) has an average 27 MGD daily water flows and a maximum of 35 MGD daily water flow.

Table 4-1: Permitted, Average, and Maximum Daily Flows

Water Treatment Plants	Permitted (MGD)	Average		Maximum	
		Flow (MGD)	Percent of Permitted	Flow (MGD)	Percent of Permitted
Toho I	54.4	27.424	50.40%	34.604	63.60%
Toho II (Harmony)	1.296	0.303	23.40%	0.729	56.30%
Toho III (Formerly)	5.262	0.127	2.40%	1.406	26.70%
Toho IV	11.743	6.6	56.20%	9.141	77.80%
Total	72.701	34.454	47.40%	45.88	63.10%

Source: TWA 2018-2022 Capital Budget

ANALYSIS OF NON-CENTRAL PUBLIC POTABLE WATER FACILITIES

For non-central public water supply systems, level of service may be expressed as minimum estimated flow guidelines used by the FDEP in the review of potable water facility permit applications. These guidelines, which consider land use and defined service areas, are used to assure that public drinking water systems provide minimum levels of service. With the exception of private wells and systems serving fewer than 25 persons and fewer than 15 connections, FDEP reviews all permit applications for public water facilities. Another concern for the City is operational problems frequently associated with this type of facility. According to FDEP officials, the problems with this type of facility result from the inability of owners to support the high costs involved in the operation of a water facility.

To ensure the health, safety and welfare of residents within the City, the City requires new private central water facilities to be designed to meet standards to facilitate connection to a public water system when they become available.

FUTURE WATER SUPPLY PLANNING EFFORTS

Planning for the protection of water resources and future potable water supplies is an important concern for the City. To protect water resources within the City, the City has established a wellhead protection area policy in its Comprehensive Plan. Regionally, the City coordinates with the TWA, nearby jurisdictions, and the SFWMD in water planning efforts.

WELLHEAD PROTECTION AREAS

The standards for wellhead protection, based on Rule 62-521.200(7), F.A.C., will assist the City in regulating land use activities and Future Land Use Map categories in the areas surrounding existing and proposed potable water supply systems. These regulations allow the City to protect areas around potable water wellheads to ensure the public health, safety, and welfare is protected.

Based on State rules, uses including solid waste disposal operations (as defined in Chapter 62-701, F.A.C.); any sanitary sewer disposal facility, excluding septic tanks; land application of reclaimed wastewater; and the disposal of sludge materials will be prohibited within 500 feet of a potable water wellhead. In addition, these rules limit the proximity of potable water wellheads to on-site sewage treatment and disposal systems and hazardous substances, as defined by the EPA. In certain situations, exemptions may be granted for utilities, continuous traffic, vehicle use, and the application of pesticides and herbicides within the wellhead protection area.

The City has entered into a contract with the County, City of St. Cloud, and SFWMD for the evaluation of the County's wellhead situation and the development of a comprehensive wellhead protection plan.

CENTRAL FLORIDA WATER INITIATIVE – REGIONAL WATER SUPPLY PLAN

The Regional Water Supply Plan is collective effort in the Central Florida Water Initiative completed by the South Florida Water Management District, St. Johns River Water Management District (SJRWMD), and Southwest Florida Water Management District (SWFWMD) in conjunction with various technical teams and stakeholders. The Central Florida Water Initiative (CFWI) was approved by the South Florida Water Management District (SFWMD) in November of 2015.

The CFWI plan projected that the total water demand in three water management districts is expected to increase by 310.95 MGD or 40 percent by the year 2035.

Table 4-2 shows the demand projection summaries for each water use category. Strategies, rules, and regulations were also established to optimize the use of water resources for the three water management districts.



Table 4-2: Summary of Projected Water Demand (MGD) in the Regional Planning Area

Category	Water Demand Projections						2035 1-in-10 Demand	2010-2035 (5-in-10) Change	2010-2035 (5-in-10) Percent Change
	2010	2015	2020	2025	2030	2035			
Public Supply									
	436.0	477.28	525.57	572.43	615.94	654.34	693.61	218.31	50%
Domestic Self-supply and Small Utilities (DSS)									
	20.36	20.22	20.75	21.92	23.13	24.42	25.89	4.06	20%
Agriculture									
	185.2	218.78	217.24	216.09	215.30	214.84	321.18	29.60	16%
Commercial/Industrial/Institutional and Mining/Dewatering (CII and MD)									
	74.05	71.47	76.74	82.82	89.29	95.85	95.85	21.80	29%
Power Generation									
	17.20	17.93	18.93	20.00	21.18	22.41	22.41	5.21	30%
Landscape/Recreational/Aesthetic (LRA)									
	40.21	44.78	51.05	57.54	64.31	72.18	89.05	31.97	80%

Source: Central Florida Water Initiative, Regional Water Supply Plan, 2015

Note: mgd = million gallons per day.

The Ten-Year Water Supply Facility Work Plan included all public, private, and regional water supply projects and programs necessary to serve the Toho Water Authority for the next 10-year period, including all development of traditional and alternative water supply sources, reuse and conservation. Table 4-3 shows each project's source of water, amount of finished water produced, the anticipated completion date, and the capital cost. The combined finished water capacity for all projects is 56-58 MGD, which is sufficient to supply the total water demand increase (32 MGD) in the area through year 2035.

Table 4-3: Water Supply Projects

Project	Implementing Agency	Finished Water Capacity (MGD)	Water Source	Estimated Completion Date	Capital Cost (Million Dollars)
Cypress Lake AWS WTP Water Main Extension to Poinciana	TWA	N/A	Exiting Source	2017	9.88
Cypress Lake AWS Reverse Osmosis WTP	TWA, St Cloud, OCU, Polk County and RCID	15-17	Brackish groundwater	2017-2021 for Phase I	131.5
Sandhill Road WRF Expansion Phase I (reuse storage tank)	TWA	4.5	Reuse	2020	1.38
City of Kissimmee West Ditch Stormwater Capture for Reuse	TWA	4.5	Stormwater Reuse	2020	10
160-Acre AWS Project - Indirect Potable Reuse	TWA	5	Rapid Infiltration Basin (RIB)	2019	14.29
Kissimmee River Basin AWS	Water Coop of Central FL	up to 25	Surface - Kissimmee River Basin - Conjunctive Use	TBD	TBD
Shingle Creek Reuse	TWA	2	Shingle Creek	2015	n/a

Source: CFWI RWSP, per TWA 5-Year Capital Budget for 2018-2022

CITY OF KISSIMMEE WATER SUPPLY WORK PLAN

The City does not have responsibility for the water supply facilities within its jurisdiction. However, Florida Statutes require the City to project its potable water supply needs for the next 10 years, which will require close coordination with TWA to ensure future potable water supplies will be available to serve future development.

Following the adoption of the CFWI plan by SFWMD, the City of Kissimmee conducted a Water Supply Work Plan to identify and plan for the water supply sources and facilities needed to serve existing and new development within the City's jurisdiction.



As a result of developing the Work Plan, the City has been able to:

- Provide potable water demand projections that will assist the City with its planning and development actions;
- Identify alternative water supply sources that will help meet future water demand needs, consistent with the Regional Water Supply Plan;
- Identify water supply capital projects to support existing and future development; and

The Water Supply Work Plan developed a level of service standard based upon gallons per day per capita for residential and non-residential uses, as shown in Table 4-4.

Table 4-4: Potable Water Supply Standards

Land Use	Toho Water Authority (TWA) Standard
Residential	96 gpd per capita
Hotel/Motel	120 gpd per room
Commercial	0.1 gpd per gross square foot of floor area
Public or Private Schools	4.8 gpd per student
All Land Use Types	98 % of the maximum
	permitted capacity and
	minimum 20 PSI pressure
	will be maintained during fire flow conditions

Source: City of Kissimmee Water Supply Work Plan, 2018

One method of offsetting the effects of withdrawals on environmental resources is to provide increased recharge to the Floridan Aquifer. An area where this occurs is in the western portion of the TWA service area, west of the City. The TWA currently uses RIBs to provide treated wastewater recharge to the Surficial and Floridan aquifers when consumptive uses of reuse water are smaller than the quantity of reuse water produced. Utilization of this recharge concept also provides the benefit of minimizing impacts to wetlands through reduced flows and surface water discharge.

WATER CONSERVATION AND REUSE PROGRAMS

Voluntary residential and commercial water conservation will be achieved through the City's participation in water conservation efforts promoted and required by the SFWMD. Locally, the City must promote the use of water-saving plumbing devices including low-flow toilets, showerheads, and faucets within new developments, as well the development and expansion of reclaimed water systems.

CONSERVATION

The City of Kissimmee is working with Toho Water Authority, Osceola County and the City of St Cloud to develop water conservation strategies, including:

- Enacted restrictions on irrigation during periods of time of critical concern that are consistent with 40E-24 of the Florida Administrative Code.
- Declared that it is unlawful to use the freshwater supply for any purpose other than cooking, drinking, or bathing during an emergency (Kissimmee Code of Ordinances 15-1-6).
- Provided a list of allowable plant species, with regulations that at least 50% of all trees and bed areas with other plant material shall be native or drought tolerant (Kissimmee Ordinance 14-2-91 (B)(8)).
- Adopted the Florida Building Code (FBC) to regulate plumbing flow.
- Implemented a number of water conservation steps in the Land Development Code.
- Continues to coordinate future water conservation efforts with the TWA and the SFWMD.

TWA conducts its own water conservation programs, and employs a full-time water conservation coordinator and two part-time conservation staff. TWA's strategies include, but not limited to, the following:

- Water conservation initiative programs, including free commercial water audits, free rain sensors and installation, free pre-rinse spray valves for restaurants, free irrigation evaluations for homeowners, and free landscaping and irrigation maintenance workshops;
- Water waste resolution enforcement; and





- In-school education and presentations to civic groups and homeowners association.
- Invested in many different tools and techniques.
- Developed new customer engagement software is included in the Authority's capital budget to help customers understand their usage in real time.
- Provided Water Conservation information on the TWA website.

REUSE

Water reuse in central Florida is accomplished primarily through the use of reclaimed water for irrigation. Reclaimed water is treated wastewater that has received at least secondary treatment and basic disinfection, and is reused after flowing out of a domestic wastewater treatment facility. Reuse is the deliberate application of reclaimed water for a beneficial purpose in compliance with the rules of FDEP and the SFWMD. Reclaimed water strategies in the regional water supply plans can include such measures as development of urban reclaimed water systems, reclaimed water system interconnections, supplemental supplies for peak flows, and aquifer storage and recovery. TWA provides reclaimed water for non-potable uses such as landscape irrigation and as cooling water for an electric-power generating plant. Within Osceola County, reclaimed water is used primarily for residential and golf course irrigation, but is also utilized for agricultural and industrial uses.

TWA operates wastewater treatment plants and reclaimed water distribution systems, including a network of 193 miles of reclaimed water distribution mains. Currently they are located along Boggy Creek Road, west of Florida's Turnpike north of Woodcrest Boulevard, and southeast of the airport along Clay Street. Reclaimed water from TWA's water reclamation facilities is either used for irrigation customers or is routed to RIBs for groundwater recharge.

As water suppliers within the Kissimmee Basin Water Supply Area are being required to increase the amount of potable water from alternative sources and to become less dependent on traditional groundwater sources, reclaimed water is looked at as an alternative to using potable water for landscape irrigation.

ALTERNATIVE WATER SUPPLIES

CENTRAL FLORIDA WATER INITIATIVE

The Central Florida Water Initiative has identified the several issues regarding water supplies. Future water demand can not be met by traditional groundwater resources alone. Groundwater withdraw within the CFWI Planning Area is approaching its limit. The expansion of withdrawals though year 2035 will create more stress on the existing areas of water resource, and will possibly damage the water resources

and related natural system. Among the 46 water bodies within the CFWI area, 10 water bodies are currently below their established Minimum Flow Levels (MFLs) of and an additional 15 water bodies are projected to fall below their established MFLs by 2035 if projected demands were to come from traditional sources.

To meet future water demand while protecting the environment and water resources, the CFWI RWSP has established various water conservation efforts, including groundwater withdrawal optimization, prevention and recovery strategies for targeted MFL water bodies, water supply project options, and water resource development project options. About 150 potential water supply project options were identified, including:

- 37 brackish/nontraditional groundwater
- 87 reclaimed water
- 17 surface water (increased from 15)
- 6 stormwater (new)
- 3 management strategies project options

These water supply project options can cumulatively meet the future water needs and provide water resource benefit.

IDENTIFIED CAPITAL IMPROVEMENTS

As the City is not a provider of potable water to its residents, capital improvements required to build and expand water supply facilities to serve existing and new development within the City are provided by TWA. As part of periodic updates to the Work Plan, the City will include a list of capital projects to be undertaken by TWA. As required, these projects are included in the Capital Improvements Element's Five-Year Schedule of Improvements. The tabulation of specific projects and their projected costs are located in the Data Inventory and Analysis of the Capital Improvement Element. TWA projected revenues related to water supply provision and programs are also located in the Capital Improvement Element.



SUMMARY

The City of Kissimmee continues to grow at a rapid pace and with this growth comes an increased demand for water services. TWA is directly responsible for the planning, acquisition, development,



operation and maintenance of water and wastewater management and delivery systems, but coordination with the City is essential. Although the City will not be involved in the distribution of potable water to consumers, it is imperative that the City coordinate with TWA to ensure that the quality of potable water is maintained and improved and that adequate supplies of potable water are available for City residents.

CHAPTER 4B: SANITARY SEWER SUB-ELEMENT

INTRODUCTION

As discussed in the Potable Water Sub-Element, the City no longer provides sanitary sewer and reuse service for its residents. Domestic and industrial wastewater services are provided through a central system operated and maintained by the Tohopekaliga Water Authority (TWA), or via on-site wastewater disposal systems, such as septic tanks and package plants, which tend to serve a variety of small developments such as residential dwelling units, mobile home parks, and small commercial sites.

At the State level, the Florida Department of Environmental Protection (FDEP) is the primary agency responsible for implementing and monitoring wastewater treatment. In addition, FDEP has adopted rules for the treatment of wastewater under Chapter 62-600, Florida Administrative Code (F.A.C.). The Florida Department of Health (FDOH) regulates on-site sewage treatment and disposal system (OSTDS) installation within the State, and works cooperatively with Osceola County through the Osceola County Health Department. OSTDS systems are regulated by Chapter 64E-6, F.A.C.

A sanitary sewer treatment system involves a collection, treatment, and distribution system. Wastewater collected through central collection systems, or sanitary sewer lines, from residences or businesses are directed to wastewater treatment plants. TWA's system within the City consists of four wastewater treatment facilities (also known as water reclamation facilities (WRFs)). The TWA system, as a whole, has over 300 wastewater pump stations, and over 200 miles of reclaimed water distribution mains, including several miles of interceptor and gravity lines, force mains and effluent disposal facilities in the City.

The TWA owns, maintains, and operates the central sanitary sewer system that serves both residential and non-residential customers in the City. The number of total wastewater customers in the TWA service area is over 71,000, and TWA treats approximately 21 million gallons of wastewater daily for their 10,000 reuse water customers.

Table 4-5 below displays the domestic wastewater facilities serving the City of Kissimmee.

Table 4-5: Central Domestic Water Reclamation Facilities serving the City, 2009

Name	Treatment Process Summary	Capacity (MGD)	DW Class	Permit Issue Date	Permit Expiration Date
TWA - South Bermuda	E-A Dual Oxidation Ditch/Flow Equalization; Reuse To Imperial & Daniels Rib Sites	8.5	IA	7/21/2003	7/17/2008
TWA - Sandhill Road WWTF	Oxidation Ditch W/ Flow Equal & Filters W/Effluent to 2mg Storage Tank and 6 Ribs	6	IIIC	5/12/2008	5/11/2013
TWA - Camelot S/D WWTF	Anox/Oxic Activated Sludge/Surge Control Filters/Reuse To Ponds/Sprayfield/G.C.	5	IA	7/16/2007	7/15/2012
TWA - Parkway WWTF	Extended Aeration WWTP W/Surge Filters, 2 Perc Ponds, Public Access Irrigation	1.5	IB	3/19/2009	3/16/2014

Source: FDEP, 2009.

Note: Domestic Wastewater Class Descriptions

IA: Advanced Wastewater Treatment (AWT): flow $\geq$ 3 MGD

IB: AWT: 0.5 MGD $\leq$ flow < 3 MGD

IC: AWT: no flow < flow < 0.5 MGD

IIIC: Act Sludge/Cont Stab: no flow < flow < 1 MGD

IIIC: Extended Aeration: 0.025 MGD $\leq$ flow < 2 MGD

IIID: Extended Aeration: no flow < flow < 0.025 MGD

Table 4-6 below identifies the design capacity, and potential capacity, of existing WWTFs serving the City. The table also indicates the current average daily flow and the reuse system's average daily flow.

Table 4-6: Wastewater Treatment Facility Capacity and Average Daily Flow

Domestic Water Reclamation Facility (WRF)	Existing Design Capacity	Current Average Daily Flow	Surplus Capacity	Potential Facility Capacity	Reuse System Average Daily Flow
South Bermuda	13.0	9.4	3.6	27.0	9.4
Sand Hill	6.0	3.5	2.5	16.0	3.4
Camelot	5.0	3.4	1.6	5.0	3.0
Parkway	1.5	0.9	0.6	2.5	0.9
TOTAL	25.5	17.2	8.5	50.5	17.2

All figures in million gallons per day (MGD).



Wastewater treatment facilities are not always connected to a central system of mains, pipes, and treatment units. These isolated facilities are of two main types: package plants, OSTDS (septic tanks). Most package treatment plants in the Kissimmee area have a treatment capacity of less than 100,000 gallons per day. Effluent is usually chlorinated and distributed to polishing or percolation ponds. Some of the larger facilities use treated effluent for irrigation purposes. Package treatment plants must be designed to a quality level that will allow for the future connection to a publicly owned central sewer system.



For these systems, wastewater disposal may be expressed in terms of minimum estimated flow guidelines used by the FDEP for reviewing sewage facility applications. These guidelines, which consider land use and defined service areas, are used to assure the provision of minimum levels of service.

On-site Sewage Treatment and Disposal Systems (OSTDS) are more commonly known as septic tanks, and are located in a few remaining areas throughout the City. Since the City does not permit new development to occur without connecting to a central sewer system, the likely locations for septic tanks would be older developments recently annexed by the City, and which are not located near a central sewer main. The soils underlying the City tend toward severe limitations for septic tank absorption fields, hence the near universality of central systems within the City and mandatory connection to them.

ANALYSIS OF FUTURE SANITARY SEWER DEMANDS AND CAPACITY

FUTURE DEMANDS

The City's demand for wastewater flow and capacity will increase over time as the population grows and development continues. Table 4-7 below uses the population projections in the Future Land Use Element, and the City's potable water Level of Service Standard yield an average daily demand. The results indicate adequate wastewater treatment capacity through the planning horizon at the South Bermuda facility.

Table 4-7: Future Wastewater Treatment Facility Capacity and Average Daily Flow Demand for the South Bermuda WWTF

Quantities	2015	2020	2025	2030	2035	2040	2045
Population	65,422	74,560	82,688	88,416	91,392	92,416	92,445
Average Daily Potable Water/Wastewater Demand (gallons per capita) ¹	96	96	96	96	96	96	96
Average Daily Demand (MGD)	6.3	7.2	7.9	8.5	8.8	8.9	8.9
South Bermuda WW Capacity (MGD)	27	27	27	27	27	27	27
Surplus Treatment Capacity	20.7	19.8	19.1	18.5	18.2	18.1	18.1

Source: TWA, 2010, Cambridge Systematics, 2018.

FACILITY NEEDS AND PLANNED IMPROVEMENTS

The TWA's wastewater treatment system for the City is projected to meet the anticipated demand for wastewater treatment capacity, and is not anticipating any improvements to facilities serving the City, at this time.

SUMMARY

The City no longer collects, treats, or disposes of wastewater, however planning and assessments of facilities are still needed. The City will continue to coordinate with the TWA, Osceola County, other local utility providers, and the FDEP to ensure adequate wastewater treatment capacity exists to serve the City, without compromising the health of residents or the natural environment. These regional partnerships will also ensure the continued utilization of reuse water for non-potable uses, which promotes the conservation of valuable groundwater resources.

Projected growth in the City of Kissimmee is not projected to outstrip TWA's ability to provide adequate treatment capacity to wastewater customers within the City. Additionally, as wastewater demand increases toward 75 percent of permitted facility capacity, TWA begins to design upgrades to the facility needed to accommodate increased flows.

CHAPTER 4C: AQUIFER RECHARGE SUB-ELEMENT

An aquifer is a geologic unit that has the ability to store and transmit water, and may be tapped with wells to provide useable water supplies, while groundwater is the water located in an aquifer. Accessed by wells, groundwater is the most common source for potable water supplies in the State.

¹ Non-residential uses are not included.



GROUNDWATER AND AQUIFERS

There are two such aquifers in the geology underlying the City: the Surficial Aquifer, and the Floridan Aquifer. The Surficial Aquifer reaches a sub-surface depth of approximately 25 feet, while the top of the Floridan Aquifer lies between 400 and 1,000 feet below the City's surface. The two geologic areas are separated by a confining layer of low-permeability sandy clay, silt, phosphorite sand, and phosphatic limestone. This layer, the Hawthorne Group, is 25 feet thick under Kissimmee, and restricts the vertical movement of water between the two aquifers.

FLORIDAN AQUIFER SYSTEM

Of the two aquifer systems present in the City, the Floridan Aquifer has greater significance, since it supplies nearly 90 percent of all groundwater used within the County for domestic, agricultural, or industrial purposes, and nearly 100 percent within the City. Water quality varies by depth and location, but the SFWMD considers the aquifer to be in good condition in the City's vicinity. Generally, as the depth increases, the level of mineralization increases, which is a symptom of low-quality potable water. The maximum water yield capacity for this aquifer is not available.

SURFICIAL AQUIFER SYSTEM

The Surficial Aquifer System underlies the entire county, but only contributes a small percentage of overall supply. In the City, this source is almost entirely unused, due to the presence of very fine-grained surficial sediments that penetrate and clog screened wells. In spite of this condition, the water quality is considered "good" by the SFWMD. Recharge of this system occurs through rainfall percolation, as well as seepage from surface waterbodies.

The City's individual domestic well users tap into the Surficial Aquifer, because the shallow depth leads to lower costs of digging wells. Some users also tap this system for irrigation use.

REGULATIONS

State and Regional

The FDEP has the responsibility for establishing and enforcing water quality standards for water supplies in the State. In implementing the Florida Safe Drinking Water Act (Chapter 403, Florida Statutes), the FDEP has developed rules classifying aquifers and regulating their use (Chapter 17-22, Florida Administrative Code). FDEP has also established guidelines for facilities that discharge to surface waters (Chapter 17-4.245, Florida Administrative Code), and which inject materials directly into groundwater (Chapter 17-28, Florida Administrative Code).

In addition to identifying the nature and extent of groundwater resources, the SFWMD implements regulatory programs for managing the State's water resources. The District regulates and manages groundwater through consumptive use permits (CUPs) for public water supplies, industry, and golf course irrigation. The District regulates and issues well construction permits for wells at least six inches in diameter. The SFWMD also has a set of water use restrictions for the conservation of water resources during times of drought. Droughts are defined by their severity, and a set of phased water restrictions is enforced according to the severity of the drought. Additionally, the SFWMD has recently mandated that all residential and commercial irrigation is only allowed twice per week.

The City and the County both have on-site stormwater retention requirements that are relevant to aquifer recharge. Additional discussion of these stormwater management levels of service are contained in the Stormwater Management Sub-Element.

GROUNDWATER RECHARGE

Groundwater recharge is the hydrogeologic process by which aquifers are replenished. Recharge occurs when water infiltrates, or percolates, through the land surface and flows into the groundwater. Recharge areas are the land surface in which infiltration occurs. Due to geologic processes, and time, a recharge area is not always located directly above the aquifer it is recharging.

The protection of recharge areas is critical in locations that rely on aquifers as the sole source of drinking water. Well-drained areas are often subject to alteration by development since these areas are typically prime land. Impervious surfaces including roads, roofs, and parking areas also reduce the area available for percolation, which reduces the rate of aquifer recharge. The quality of the water percolating into the aquifer may in turn compromise groundwater quality. Soils absorb contaminants, which may be transmitted via percolation into the groundwater supply, impacting potable water resources. Solid waste and hazardous waste facilities, underground storage tanks, and septic systems all have the potential to contaminate groundwater sources.

The City is generally located within a moderate to poor recharge area, and tends to only contribute less than four inches of recharge per year. The City does not have any areas with high rates of groundwater percolation. These areas exist further to the west in Osceola County.

PROTECTION OF GROUNDWATER RESOURCES

The Floridan Aquifer System provides a supply of freshwater, including the source for the Tohopekaliga Water Authority (TWA) serving the City. For those on individual wells, low-to-moderate yields are





provided from the Surficial Aquifer. Aquifer contamination is most likely in areas with high rates of groundwater recharge, which the City does not have. Regardless, this condition is due to the relative thinness of the confining layer above the Floridan aquifer.

Conservation of groundwater supplies are critical, and will continue to remain so, in the maintenance of not only potable water supplies, but also the maintenance of surface water flows and water levels within environmentally sensitive wetland habitats. The City coordinates with Osceola County, TWA, the SFWMD, and the FDEP regarding water conservation measures and water supply demand.

SUMMARY

The protection of groundwater resources is directly linked to the regulation of stormwater, and the conservation of potable water resources. Additional information regarding both groundwater supplies, and the protection of groundwater quality through stormwater management, can be found in the Potable Water Sub-Element and the Stormwater Management Sub-Element, respectively.

The City's anticipated future development pattern with more mixed uses and higher densities has the potential to reduce the City's overall per-capita consumption of groundwater, by allowing greater recharge through a reduction in impervious surfaces required of typical suburban development patterns.

CHAPTER 4D: STORMWATER MANAGEMENT SUB-ELEMENT

INTRODUCTION

Water flowing overland, and into and through natural and constructed conveyance systems during and immediately following a storm event is called stormwater runoff. Stormwater management refers to the comprehensive strategies for dealing with the effects of changes in total runoff, peak flow, and water quality. Stormwater management is important to address flood prevention and protection, as well as pollutant control. Stormwater carries pollutants that are washed from impervious surfaces during rainfall, and may include bacteria, fertilizers, motor oil, metals, and trash. The central premise of managing stormwater is to ensure that the hydrology of an area after development is similar to that present before the development occurs.

Stormwater regulations are established in the Comprehensive Plan and the Land Development Code, and the City must coordinate with the EPA, the FDEP, the Department of Economic Opportunity (DEO), the SFWMD, and the County. Section 402 of the Clean Water Act, and Chapters 17-40, 62-25, and 62-40 of the Florida Administrative Code (F.A.C.) form the regulatory framework of those agencies. It is through these regulatory mechanisms that the City is able to address flooding prevention and the protection of water resources and wildlife habitat that stormwater management may provide the community. The City's Land Development Code includes numerous ordinances addressing open space, tree protection, landscaping, flood damage prevention, and reduction of flood hazards.



The City's Stormwater Division maintains structures designed to collect, convey, hold, divert, or discharge stormwater. This includes storm sewers, canals, ditches, culverts, detention structures, and retention structures. Stormwater Division crews routinely clean stormwater inlets, pipes, and mow retention/detention areas to help ensure the City's stormwater management system runs properly.

All of the rain that falls in the City of Kissimmee either goes into the ground or into the City's stormwater system. The rainwater enters the stormwater system through inlets and is transported to retention/detention ponds by a network of swales and pipes. While the stormwater is in the ponds, heavy metals and suspended sediment fall out of the water. The stormwater is discharged once the pond reaches a certain level. The water that leaves the pond goes through a series of pipes until it reaches one of the main ditches/waterways. These then discharge the stormwater and any remaining pollution into Lake Tohopekaliga.

There are six watersheds located within the City, including Shingle Creek, Mill Slough, Bass Slough, East City Ditch, West City Ditch, and Downtown. The Tohopekaliga Basin (part of the Upper Kissimmee River Basin) is the basin into which all surface waters in the City eventually flow.

NATURAL DRAINAGE WAYS

The City is located in the Upper Kissimmee River Basin, which is under the jurisdiction of the SFWMD. The "urban" portion of the basin in which the City is located extends generally from the southern shore of Lake Tohopekaliga north to the County line.

Dredge and fill activities have significantly altered the original hydrography of the area. Soils in the City are primarily associated with high water tables, and these are coupled with a weakly cemented sand layer about 1.5 to 3 feet deep. This combination of soils in the City means that surface water tends not to percolate quickly. The *Soils Map* identifies the soil types located in and around the City, and the *Floodplain and Topography Map* demonstrates the gentle slope of topography from the northern part of the City toward Lake Tohopekaliga. These maps can be found in the Conservation Element.

Lake Tohopekaliga

This large, shallow lake on the City's southern border covers 21,400 acres and receives inflow primarily from Shingle Creek to the north, and from East Lake Tohopekaliga via the St. Cloud Canal to the east. The lake flows out southward through the Kissimmee Chain of Lakes, which eventually flows into Lake Okeechobee and into the Everglades. The Lake was altered as early as the 1880s with the construction of several canals to link nearby lakes. Other inflows to the Lake have been channelized, and the lake now receives about one-third of its total flow through channelized streams and ditches. Shingle Creek, the St. Cloud Canal, and rainfall contribute the majority of the flow, with the remainder flowing from minor basins and groundwater seepage.



Mill Slough

Mill Slough is a natural drainage system with a watershed area of 11.6 miles. Its northernmost reach is in Orange County, and flows south through developed areas of Osceola County and the eastern portion of City before flowing into Lake Tohopekalliga. Of the total watershed, about 20 percent was developed prior to the regulation of stormwater discharge, which began in the 1980s. A study in the late 1980s identified deficiencies in the City's stormwater infrastructure in this area, and the City responded with increased investment in stormwater conveyances and structures to better manage stormwater runoff and flood control in the areas around Mill Slough.

Shingle Creek

Considered the headwaters of the Everglades, Shingle Creek runs through the western portion of the City through Osceola County from Orange County south to Lake Tohopekalliga. The Creek's headwaters are in Orange County, and the waters flow south along the western edge of the City. The Shingle Creek watershed within unincorporated County areas and the City is 90 percent undeveloped.



In the past, the Creek was found to have high concentrations of nutrients, primarily resulting from effluent discharge from local sewage treatment plants. These effluent discharges have been eliminated, and no point sources of pollution exist along the Creek. The water is, however, still susceptible to urban stormwater runoff from both the City, and further upstream in Orange County. The basin includes both the US 192 and US 17-92 corridors, and those roadways' stormwater conveyances. Presently, as developments are proposed within the City, stormwater attenuation and water quality are controlled as each subdivision or commercial property is constructed, including interconnected ponds to slow water flow, and reduce nutrient loads.

The City, the County, Orange County, and the SFWMD have been coordinating on programs to protect the Creek. The SFWMD has an adopted management plan for Shingle Creek that outlines the District's management goals, which include restoration of hydrological and ecological characteristics, exotic plant control, and public access for active and passive recreational uses. Portions of the Creek that run through the City are buffered by protected Open Space land uses. In 2008, the City partnered with the County and Florida Communities Trust to acquire additional land for the Shingle Creek Recreational Preserve. As part of the project, a stormwater retention pond is planned, as are recreational improvements, including active and passive uses.

CITY STORMWATER DRAINAGES AND FACILITIES

The City's location adjacent to Lake Tohopekaliga and Shingle Creek requires a system of stormwater conveyances to retain runoff during storm events. These are designed to prevent immediate discharge to surface waters. The remaining four watersheds are man-made and include Bass Slough, the East City Ditch, the West City Ditch, and the downtown culvert system. The City also maintains stormwater monitoring stations throughout the City.

STORMWATER QUALITY AND QUANTITY

As storm water flows over driveways, lawns, and sidewalks, it picks up debris, chemicals, dirt, and other pollutants. Stormwater can flow into a storm sewer system or directly to a lake, stream, river, creek, or wetland. Anything that enters a storm sewer system is discharged untreated into the water bodies we use for swimming, fishing, and other recreational activities. Reducing pollutant levels, including from household chemicals and sediment will improve the quality of lakes and the overall health of the ecosystem. Several principal stormwater quality problems are identified in Table 4-8 below.

Table 4-8: Stormwater Quality Concerns, Impacts, and Causes

Issue	Negative Impacts	Causes
Sediment Accumulation	Smothering of plants and animals that live on water body bottoms	Erosion; Washing cars in the road; Dumping garbage in storm inlets; Air Pollution
Turbidity in Water	Reduces plant growth and aesthetic beauty; Fish kills	Erosion; Washing cars in the road; Air pollution in the rainwater
Nutrient Enrichment	Increase in Algae and invasive aquatic plants	Decaying plants; Fertilizers; Phosphorous in car soap; Animal feces; Illicit discharge
Gross Pollutants	Reduces plant growth and aesthetic beauty; Fish kills	Littering; Illegal dumping
Heavy Metals	Stresses aquatic plants and animals; Contamination of the food chain	Runoff from roads; Swimming pool water; rusting metal; Illicit discharge
Petrol, Oils and Grease	Reduces aquatic life and aesthetic beauty	Leaks from vehicles; Maintenance of cars; Illegal dumping; Illicit discharge
High Runoff Rates	Erosion of banks; effects wetlands; causes disturbances to aquatic ecosystem	Impervious surfaces (roads, roofs, etc.) directly connected to the stormwater system

Source: City of Kissimmee, 2009.

Stormwater quality is regulated by the FDEP, specifically through Chapter 64-25 of the Florida Administrative Code.

NATIONAL POLLUTION DISCHARGE ELIMINATION SYSTEM (NPDES)

The National Pollution Discharge Elimination System (NPDES) program was established as the fundamental regulatory mechanism of the Clean Water Act (CWA). The CWA is the primary federal



legislation that protects surface waters, such as lakes, rivers, and coastal areas. The initial focus of the NPDES program was controlling targeted point sources, such as the direct sewer outfall that used to flow into Lake Tohopekaliga. However, studies indicated that stormwater runoff from non-point sources such as agricultural, urban areas, construction sites, land disposal, and mining areas were the leading cause of water quality impairment.

The City of Kissimmee is a regulated small municipal separate storm sewer system (MS4s) and was required to develop and implement a NPDES program by September 2, 2008, and the City has complied with this requirement. The City's Code of Ordinances and policies within the Comprehensive Plan meet the NPDES requirements for stormwater quality and runoff control, and for maintenance of stormwater systems. In addition, the City has also passed ordinances that prohibit illegal dumping and illicit discharges to the stormwater system, as required by federal regulations. To further meet the federal requirement for public education and involvement, the City's education programs help enhance public awareness of the importance of minimizing pollution to stormwater management systems.

LOW IMPACT DEVELOPMENT TECHNIQUES

Closely associated with the Green Building movement is the use of low-impact development techniques for development sites, which seek to maintain the existing drainage and ecosystem functions of a site. These techniques are primarily undertaken to mimic natural stormwater runoff quantity, and improve runoff quality to local surface water bodies and areas with high rates of groundwater recharge.

Reducing pollutant levels, including from household chemicals and sediment will improve the quality of lakes and the overall health of the ecosystem.

The techniques may be structural or non-structural, and may be utilized in already developed urban areas or on sensitive sites. By taking into account the existing characteristics of a site upon which a structure is to be built, and potentially clustering impervious areas together where most appropriate, landowners and developers can contribute to resource preservation through the integration of the natural environment and the built environment, and maintain predevelopment watershed and ecological functions.

Other techniques include maintaining buffer zones and natural drainage courses, using infiltration swales, reducing impervious surfaces, and conserving open space. The use of green roofs can reduce the amount of stormwater runoff on a site, reduce the need for irrigation, and also help to insulate the roof and building against radiant heat gain. Rain gardens and constructed wetlands on a site allow for increased infiltration of water through a reduction in impervious surfaces; provide esthetically pleasing and functional landscaping that reduces the need for irrigation; and reduce impervious surfaces, contributing to a reduction in the creation of urban heat islands while also improving the quality of surface

water runoff. Green roofs, in particular, could be a solution to the stormwater mitigation problems experienced within the City's historic downtown area.

LEVEL OF SERVICE STANDARDS

Where permits are required by the County, State, or federal agencies, the City first requires a development applicant to provide a copy of the relevant permits prior to the approval of any construction activities. The City's adopted level of service standards for stormwater quantity, which serve as the basis for determining the availability of facility capacity and demand, are as follows:

- Limit the allowable stormwater peak discharge detained from a site to be developed or altered to pre-development or pre-alteration peak discharge for the 10-year frequency, 72 hour duration storm. Retention/detention ponds, for all sites greater than ten acres total area, to have a minimum of one foot freeboard.
- Limit velocities in any new system to a maximum of 3 feet/second or less for the 10-year storm.
- Roadway stormwater management system shall be designed for the 10-year storm event, utilizing the Florida Department of Transportation Rainfall Intensity Curves.
- The design storm frequency to be utilized for the design of pavement drainage shall be the 10 year.
- Water Quality - Wet detention for the first inch of runoff or the total runoff of 2.5 inches times the percent of imperviousness, whichever is greater. Reductions in the above volumes of 25% and 50% for dry detention and retention respectively can be applied when conditions permit. Industrial zoned projects shall provide at least 1/2 inch dry detention or retention pretreatment as part of the required retention/ detention.
- Incorporate a floodplain management standard which requires that no net reduction occur in flood storage (e.g., 100-year flood) for any development within the impacted area.
- The Land Development Code shall continue to include mandatory retention and/or detention of stormwater discharge from developed or altered sites, per Chapter 62-25, Florida Administrative Code.

The LOS standards above are based on the development regulations within the Land Development Code. The City also enforces those levels of standards set by the SFWMD for individual stormwater retention and detention facilities, and will abide by total maximum daily load criteria, upon their identification for Lake Tohopekaliga and Shingle Creek.



PROJECTS TO MAINTAIN LEVEL OF SERVICE STANDARDS

Stormwater projects being implemented to maintain level of service standards include improvements to the Baffle Boxes, Master Stormwater Pond, Lavon/Hill Drainage Improvements, and Woodside Drainage Improvements, all of which are funded by the *City of Kissimmee 2017-2021 Five Year Approved Capital Budget (2017)* shown in Table 8-2 of the Capital Improvement Element.

COMPREHENSIVE APPROACHES TO STORMWATER MANAGEMENT

In many cases watersheds extend into several governmental jurisdictions, and cooperation between local governments, especially through the use of interlocal agreements, can lead to successful achievement of common stormwater management goals. A comprehensive approach offers advantages over the site-specific approach contained in most regulations for land development, but it requires sufficient data and analysis of an entire watershed to identify the most appropriate control measures and optimum locations to control watershed-wide impacts.

The Downtown Stormwater Master Plan (Figure 4-1) encompasses much of the CRA in the City's downtown area. The Plan will ensure stormwater management is provided, but will not be required on each site as a matter of LOS and concurrency, as long as the entire area is meeting its determined LOS standard. Development and redevelopment projects will have the opportunity to contribute to the fund to implement the projects identified in the Master Plan, and may have the opportunity to reduce their stormwater loads through the use of low-impact development techniques.

FUNDING FOR STORMWATER MANAGEMENT

The City's Stormwater Utility Fund is assessed to City residents and businesses, and was created to pay for stormwater maintenance and improvement projects within the City. The Stormwater Operations portion funds the maintenance of City ponds, culverts, ditches, and other drainage areas. Although maintenance of privately maintained ponds and ditches in subdivisions or businesses is not included, the outfalls and spillways for these must be maintained and are subject to the fee. This also includes the sweeping of all City streets to remove sediments that would otherwise find their way into the lake system and cause possible environmental issues.

SUMMARY

The condition of the facilities is generally good, and the City's Stormwater Management Division has an ongoing schedule of maintenance. Generally, the expected life of facilities is approximately 50 years, but with regular maintenance that can be extended indefinitely. However, expansion of facilities will largely be governed by growth within the City, the areas in which the growth occurs, and the extent of plans for master stormwater systems within the City.

Projects completed, including improvements to existing conveyances, and facility upgrades associated with road construction activities, as well as enforcement of existing regulations not only manage existing conditions, they also serve as components of the City's overall system that will be in place for decades.

CHAPTER 4E: SOLID WASTE SUB-ELEMENT

This sub-element details the existing facilities servicing the City and provides data and analysis regarding future needs for solid waste collection and disposal.

INTRODUCTION

The City's Department of Sanitation is responsible for waste collection and disposal activities within the City.

WASTE COLLECTION AND DISPOSAL

The City does not operate or maintain disposal facilities or transfer stations for solid waste. The City of Kissimmee operates and maintains solid waste collection services for single-family residential uses. Only City residents are provided service by the City's collection system. Non-residential and multifamily residential collection services are supplied by a vendor through an exclusive franchise agreement with the City. They collect solid wastes from commercial dumpsters and "roll-offs" within the City limits.

Since 2011, the City has provided curb-side collection of recyclable materials. The Osceola County Solid Waste Office also encourages recycling efforts by providing recycling drop-off facilities at four locations without charge to County residents.

All municipal solid waste (MSW) collected in the City (residential and non-residential) is transferred to the J.E.D. Solid Waste Facility. Prior to arrival at the landfill, MSW is tipped at the City of St. Cloud Transfer Station. Although hazardous materials are not collected by the City, a public drop-off is located at the Bass Road Facility, which is also a site of a drop-off collection center for household recyclable materials. Household chemical wastes are also collected at the City of St. Cloud Transfer Station.

Completed in 2004, a privately owned and maintained Class I Solid Waste Disposal Landfill, the J.E.D. Solid Waste Facility in Holopaw, is capable of servicing MSW disposal needs for all county residents for 30 years (2034), based on current rates of waste disposal. (See below for discussion of levels of service). Land is reserved for supporting facilities and required buffer areas, and the maximum height of the landfill is limited to 330 feet. Hazardous wastes are not accepted at this facility, but construction and demolition debris is accepted.



FUTURE DEMAND AND CAPACITY ASSESSMENT

As noted in the Future Land Use Element data inventory and analysis, the projected population in 2045 will be 92,445 permanent residents. Table 4-9 below identifies the projected rates of MSW production within the City, using the 2008 rate of 2.7 pounds per capita per day, and the projected population figures.

Table 4-9: Projected Rates of Residential Solid Waste Production

	2015	2020	2025	2030	2035	2040	2045
Projected Population	65,422	74,560	82,688	88,416	91,392	92,416	92,445
Waste Generated (in tons)	130,844	149,120	165,376	176,832	182,784	184,832	184,890

The data indicates that if the rate of waste generation remains the same, the total quantity of waste generated in a year will remain within the threshold set by the adopted level of service standard of 5.1 pounds per capita per day.

The projected demand takes into account the production of wastes from residential (single family) uses only. Non-residential wastes are not collected by the City, and data regarding their rates of waste production are not available.

LEVEL OF SERVICE STANDARD

The City currently has an adopted level of service of 5.1 pounds per capita per day. The City does not currently have an adopted level of service standard for a rate of waste generation for non-residential uses. This does, however, mean that a method by which to track non-residential waste production should be established.

SUMMARY

In addition to the coordination of level of service standards, the City will continue to encourage the recycling of household recyclables to reduce the City's landfill-bound waste stream. The City will also continue to coordinate with the County and private hauling companies to monitor the generation of waste and methods by which to reduce rates of waste production and burial, should the need arise.